EDS Lab Assignments

Practical 3

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3.1.1 Numpy Array Operations

Problem Statement:
Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers r and c, representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Code:

```
import numpy as np
rows, cols = map(int, input().split())
elements = []

for _ in range (rows):
        elements.extend(map(int, input().split()))

array = np.array(elements).reshape(rows, cols)
print(array)
print(array.ndim)
print(array.shape)
print(array.size)
```

Output:

Average time	Maximum time
0.039 s	0.056 s
39.00 ms	56.00 ms

2 out of 2 shown test case(s) passed

10 out of 10 hidden test case(s) passed

Test case 1 48 ms Debug

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]	[[1 2 3 4]
[5 6 7 8]	[5 6 7 8]
[9 10 11 12]]	[9 10 11 12]]
2	2
(3, 4)	(3, 4)
12	12

Test case 2 14 ms Debug

Expected output	Actual output
0 0	0 0
[]	[]
2	2
(0, 0)	(0, 0)
0	0

3.2.1 Numpy: Matrix Operations

Problem Statement:

The given code takes two 3×3 matrices, matrix_a, and matrix_b, as input from the user and converts them into NumPy arrays.

Task:
You are required to compute and display the results of the following matrix operations:

1. **Addition** (matrix_a + matrix_b)
2. **Subtraction** (matrix_a – matrix_b)
3. **Element-wise Multiplication** (matrix_a * matrix_b)
4. **Matrix Multiplication** (matrix_a · matrix_b)
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for matrix_a, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.

Output Format:
The program should display the results of the operations in the following order:

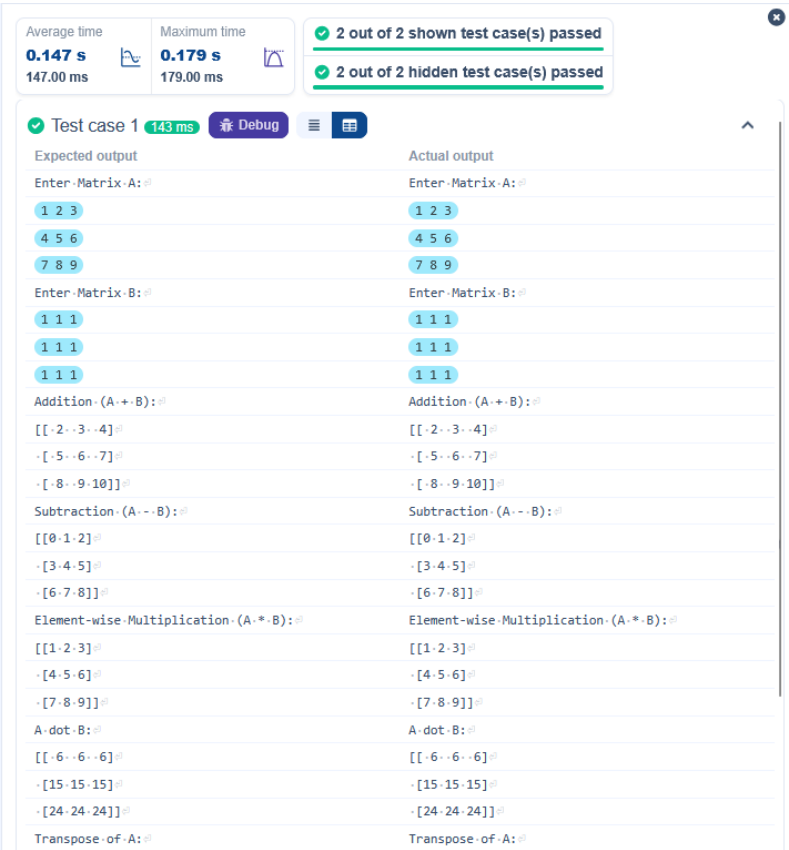
1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Code:

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition
addition_result = matrix_a + matrix_b
print("Addition (A + B):")
print(addition_result)

# Subtraction
subtraction_result = matrix_a - matrix_b
print("Subtraction (A - B):")
```

```
print(subtraction_result)

# Multiplication (element-wise)
elementwise_multiplication_result = matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result)

# Matrix multiplication (dot product)
matrix_multiplication_result = np.dot(matrix_a, matrix_b)
print("A dot B:")
print(matrix_multiplication_result)

# Transpose
transpose_a = matrix_a.T
print("Transpose of A:")
print(transpose_a)
```

Output:



Average time	Maximum time
0.147 s	0.179 s
147.00 ms	179.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Transpose of A:	Transpose of A:
[[1 4 7]	[[1 4 7]
[2 5 8]	[2 5 8]
[3 6 9]]	[3 6 9]]

Test case 2 141 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
2 4 8	2 4 8
9 6 5	9 6 5
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
10 12 13	10 12 13
14 15 16	14 15 16
17 18 19	17 18 19
Addition (A + B):	Addition (A + B):
[[12 16 21]	[[12 16 21]
[23 21 21]	[23 21 21]
[24 26 28]]	[24 26 28]]
Subtraction (A - B):	Subtraction (A - B):
[[-8 -8 -5]	[[-8 -8 -5]
[-5 -9 -11]	[-5 -9 -11]
[-10 -10 -10]]	[-10 -10 -10]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[20 48 104]	[[20 48 104]
[126 90 80]	[126 90 80]
[119 144 171]]	[119 144 171]]

A dot B:	A dot B:
[[212 228 242]	[[212 228 242]
[259 288 308]	[259 288 308]
[335 366 390]]	[335 366 390]]
Transpose of A:	Transpose of A:
[[2 9 7]	[[2 9 7]
[4 6 8]	[4 6 8]
[8 5 9]]	[8 5 9]]

3.2.2 Numpy: Horizontal and Vertical Stacking of Arrays

Problem Statement:

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)
a = np.hstack((arr1, arr2))
print("Horizontal Stack:")
print(a)

# Perform vertical stacking (vstack)
b = np.vstack((arr1, arr2))
print("Vertical Stack:")
print(b)
```

Output:

Average time	Maximum time
0.150 s 149.75 ms	0.171 s 171.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 111 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6
7 8 9	7 8 9
10 11 12	10 11 12
Horizontal Stack:	Horizontal Stack:
[[1 2 3 4 5 6]	[[1 2 3 4 5 6]
[4 5 6 7 8 9]	[4 5 6 7 8 9]
[7 8 9 10 11 12]]	[7 8 9 10 11 12]]
Vertical Stack:	Vertical Stack:
[[1 2 3]	[[1 2 3]
[4 5 6]	[4 5 6]
[7 8 9]	[7 8 9]
[4 5 6]	[4 5 6]
[7 8 9]	[7 8 9]
[10 11 12]]	[10 11 12]]

Test case 2 171 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
-1 -2 -3	-1 -2 -3
-4 -5 -6	-4 -5 -6
-7 -8 -9	-7 -8 -9
Enter Array2:	Enter Array2:
10 11 12	10 11 12
13 14 15	13 14 15
16 17 18	16 17 18
Horizontal Stack:	Horizontal Stack:
[[-1 -2 -3 10 11 12]	[[-1 -2 -3 10 11 12]
[-4 -5 -6 13 14 15]	[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]]	[-7 -8 -9 16 17 18]]
Vertical Stack:	Vertical Stack:
[[-1 -2 -3]	[[-1 -2 -3]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[10 11 12]	[10 11 12]
[13 14 15]	[13 14 15]
[16 17 18]]	[16 17 18]]

3.2.3 Numpy: Custom Sequence Generation

Problem Statement:

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Code:

```
import numpy as np
# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())
# Generate the sequence using np.arange()
a = np.arange(start, stop, step)
# Print the generated sequence
print(a)
```

Output:

3.2.4 Numpy: Arithmetic and Statistical Operations, Mathematical Operations and Bitwise Operators

Problem Statement:

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code:

```
import numpy as np

def array_operations(A, B):
        # Convert A and B to NumPy arrays
        A = np.array(A)
        B = np.array(B)

        # Arithmetic Operations
        sum_result = A + B
        diff_result = A - B
        prod_result = A * B

        # Statistical Operations
        mean_A = np.mean(A)
        median_A = np.median(A)
        std_dev_A = np.std(A)

        # Bitwise Operations
        and_result = A & B
        or_result = A | B
```

```
        xor_result = A ^ B

    # Output results with one space between each element
        print("Element-wise Sum:", ' '.join(map(str, sum_result)))
        print("Element-wise Difference:", ' '.join(map(str, diff_result)))
        print("Element-wise Product:", ' '.join(map(str, prod_result)))

        print(f"Mean of A: {mean_A}")
        print(f"Median of A: {median_A}")
        print(f"Standard Deviation of A: {std_dev_A}")
            print("Bitwise AND:", ' '.join(map(str, and_result)))
        print("Bitwise OR:", ' '.join(map(str, or_result)))
        print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split()))  # Elements of array A
B = list(map(int, input().split()))  # Elements of array B
array_operations(A, B)
```

Output:

3.2.5 Numpy: Copying and Viewing Arrays

Problem Statement:

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:
  Original array after modifying view: <**original_array**>
  View array: <**view_array**>
- After modifying the copy:
  Original array after modifying copy: <**original_array**>
  Copy array: <**copy_array**>

Code:

```
import numpy as np
inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()

# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```

Output:

3.2.6 Searching, Sorting, Counting, Broadcasting

Problem Statement:

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.
Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:
1. Searching: Find the indices where search_value appears in array1 and print these indices.
2. Counting: Count how many times count_value appears in array1 and print the count.
3. Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
4. Sorting: Sort array1 in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

Output Format:

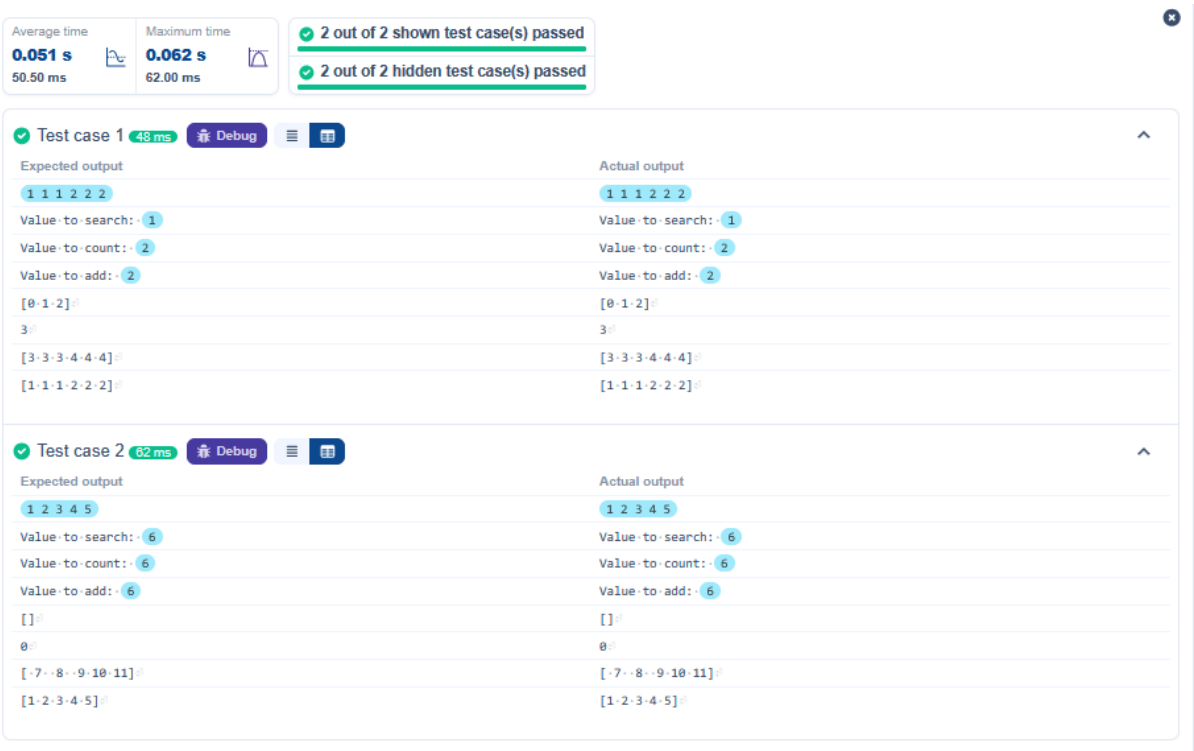
1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

Code:

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1
a = np.where(array1 == search_value)[0]
print(a)

# Count occurrences in array1
b = np.count_nonzero(array1 == count_value)
print(b)
```

```
# Broadcasting addition
c = array1 + broadcast_value
print(c)

# Sort the first array
d = np.sort(array1)
print(d)
```

Output:



3.2.7 Student Data Analysis and Operations

Problem Statement:

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.

- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1**: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Code:

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

roll = a[:, 0].astype(float)
s1 = a[:, 1]
s2 = a[:, 2]
s3 = a[:, 3]

# 1. Print all student details
print("All student Details:\n", a)

# 2. print total students
print("Total Students:", len(a))

# 3. Print all student Roll numbers
print("All Student Roll Nos", roll)

# 4. Print subject 1 marks
print("Subject 1 Marks", s1)

# 5. print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(s2))

# 6. print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(s3))

# 7. Print All subject marks
print("All subject marks:", a[:, 1:])

# 8. print Total marks of students
total = s1 + s2 + s3
print("Total Marks", total)

# 9. print average marks of each student
print(np.round(total/3, 1))

# 10. print average marks of each subject
print("Average Marks of each subject", np.mean(a[:, 1:], axis = 0))

# 11. print average marks of S1 and S2
print("Average Marks of S1 and S2", np.mean([s1, s2], axis = 1))

# 12. print average marks of S1 and S3
print("Average Marks of S1 and S3", np.mean([s1, s3], axis = 1))

# 13. print Roll number who got maximum marks in Subject 3
print("Roll no who got maximum marks in Subject 3", float(roll[np.argmax(s3)]))

# 14. print Roll number who got minimum marks in Subject 2
```

```
print("Roll no who got minimum marks in Subject 2", float(roll[np.argmin(s2)]))

# 15. print Roll number who got 24 marks in Subject 2
print("Roll no who got 24 marks in Subject 2",roll[s2 == 24].astype(float).reshape(1, -
1))

# 16. print count of students who got marks in Subject 1 < 40
print("Count of students who got marks in Subject 1 < 40", np.sum(s1 < 40))

# 17. print count of students who got marks in Subject 2 > 90
print("Count of students who got marks in Subject 2 > 90:", np.sum(s2 > 90))

# 18. print count of students in each subject who got marks >= 90
print("Count of students in each subject who got marks >= 90:", np.sum(a[:, 1:] >=
90, axis = 0))

# 19. print count of subjects in which each student got marks >= 90
print("Roll no:", roll)
print("Count of subjects in which student got marks >= 90:", np.sum(a[:, 1:] >= 90,
axis = 1))

# 20. Print S1 marks in ascending order
print(np.sort(s1))

# 21. Print S1 marks >= 50 and <= 90
print(a[(s1 >= 50) & (s1 <= 90)].astype(float))
print(a.astype(float))

# 22. Print the index position of marks 79
print(np.where(s1 == 79))
```

Output:

Average time	Maximum time
0.019 s 19.00 ms	0.019 s 19.00 ms

1 out of 1 shown test case(s) passed

Test case 1 19 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301, 67, 77, 88]	[[301, 67, 77, 88]
[302, 78, 88, 77]	[302, 78, 88, 77]
[303, 45, 56, 89]	[303, 45, 56, 89]
[304, 88, 98, 45]	[304, 88, 98, 45]
[305, 78, 88, 99]	[305, 78, 88, 99]
[306, 68, 78, 36]	[306, 68, 78, 36]
[307, 79, 12, 45]	[307, 79, 12, 45]
[308, 46, 45, 77]	[308, 46, 45, 77]
[309, 89, 32, 88]	[309, 89, 32, 88]
[310, 79, 24, 33]]	[310, 79, 24, 33]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]	All Student Roll Nos [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]
Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79.]	Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79.]
Min marks in Subject 2 12.0	Min marks in Subject 2 12.0
Max marks in Subject 3 99.0	Max marks in Subject 3 99.0
All subject marks: [[67, 77, 88]	All subject marks: [[67, 77, 88]
[78, 88, 77]	[78, 88, 77]
[45, 56, 89]	[45, 56, 89]
[88, 98, 45]	[88, 98, 45]
[78, 88, 99]	[78, 88, 99]
[68, 78, 36]	[68, 78, 36]
[79, 12, 45]	[79, 12, 45]
[46, 45, 77]	[46, 45, 77]
[89, 32, 88]	[89, 32, 88]

Total Marks [232, 243, 190, 231, 265, 182, 136, 168, 209, 136.]	Total Marks [232, 243, 190, 231, 265, 182, 136, 168, 209, 136.]
[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]	Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]	Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]	Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0	Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0	Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]	Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0	Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1	Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]	Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]	Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]	Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0 0]
[45, 46, 67, 68, 78, 78, 79, 79, 88, 89.]	[45, 46, 67, 68, 78, 78, 79, 79, 88, 89.]
[[301, 67, 77, 88.]	[[301, 67, 77, 88.]
[302, 78, 88, 77.]	[302, 78, 88, 77.]
[304, 88, 98, 45.]	[304, 88, 98, 45.]
[305, 78, 88, 99.]	[305, 78, 88, 99.]
[306, 68, 78, 36.]	[306, 68, 78, 36.]
[307, 79, 12, 45.]	[307, 79, 12, 45.]
[309, 89, 32, 88.]	[309, 89, 32, 88.]
[310, 79, 24, 33.]]	[310, 79, 24, 33.]]
[[301, 67, 77, 88.]	[[301, 67, 77, 88.]
[302, 78, 88, 77.]	[302, 78, 88, 77.]
[303, 45, 56, 89.]	[303, 45, 56, 89.]
[304, 88, 98, 45.]	[304, 88, 98, 45.]
[305, 78, 88, 99.]	[305, 78, 88, 99.]

```
 [303.  45.  56.  89.]
 [304.  88.  98.  45.]
 [305.  78.  88.  99.]
 [306.  68.  78.  36.]
 [307.  79.  12.  45.]
 [308.  46.  45.  77.]
 [309.  89.  32.  88.]
 [310.  79.  24.  33.]]
(array([6, 9], dtype=int32),)
```

```
 [303.  45.  56.  89.]
 [304.  88.  98.  45.]
 [305.  78.  88.  99.]
 [306.  68.  78.  36.]
 [307.  79.  12.  45.]
 [308.  46.  45.  77.]
 [309.  89.  32.  88.]
 [310.  79.  24.  33.]]
(array([6, 9], dtype=int32),)
```